MONSOONS

More extreme Indian monsoon rainfall in El Niño summers

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Extreme rainfall during the Indian summer monsoon can be destructive and deadly to the world's third-largest economy and most populous country. Although El Niño events in the equatorial Pacific are known to suppress total summer rainfall throughout India, we show using observational data spanning 1901 to 2020 that, counterintuitively, they simultaneously intensify extreme daily rainfall. This is partly driven by increases in extreme daily values of convective buoyancy, provided that both the undilute instability of near-surface air and the dilution by mixing with drier air above are considered. El Niño could plausibly drive similar changes in other tropical regions, and our framework could be further applied to changes in hourly extremes, to other internal variability modes, and to forced trends under climate change.

The influence of the El Niño–Southern Oscillation (ENSO) on total summer rainfall in India has been recognized since the early 20th century (*1*–*4*). Anomalously warm sea surface temperatures (SSTs) in the equatorial Pacific during El Niño summers generate anomalous season-mean ascent of tropospheric air locally but compensating descent over much of the rest of the tropics, including India.

One might expect summer-mean subsidence to inhibit moist convection of all intensities, extremes included, reducing both the mean and variance of daily rainfall. Indeed, recent studies (*5*, *6*) establish that El Niño reduces the average accumulation of precipitation on rainy days—the rainfall intensity—within southeastern India and in Rajasthan state in the northwest. These are the summer monsoon's climatologically dry areas, and within them during El Niño summers it tends to rain even less overall, less often, and less intensely when rain does fall (*5*, *6*).

But the opposite rainfall intensity signal emerges in the summer monsoon's climatologically rainy areas (*5*, *6*)—the broad Central Monsoon Zone in central India and the narrow southwestern band between the Arabian Sea and the Western Ghats mountains of peninsular India. Within them it tends to rain less overall and less often during El Niño summers but more intensely when the rain comes.

Whether these signals in rainfall intensity stem from destructive extreme daily rainfall events (*7*) or agriculturally beneficial moderate rainfall days (*8*), and what their mechanistic origins are, remain unanswered. Observational studies have investigated ENSO's influence on daily or hourly rainfall extremes in other tropical domains (*9*–*16*) but not in the Indian summer monsoon.

By comparison, the possibility of long-term positive trends in Indian summer monsoon extreme rainfall has received considerable attention (*17*, *18*), but the existence of such trends is contested (*19*), they can be sensitive to the particular metric used to characterize an extreme event (*19*), and they have largely flattened or reversed over the most recent years (*20*). Climatologically, daily rainfall in the Indian summer monsoon is strongly influenced by both intraseasonal modes as manifested in active and break periods (*21*) and by synoptic disturbances, particularly monsoon low-pressure systems (*22*, *23*). But ENSO's influence on intraseasonal modes is strongly contested (*21*, *24*), and its influence on low-pressure systems appears modest, at least on an India-wide basis (*22*). It is well established, however, no matter what the states of secular trends, intraseasonal modes, or synoptic systems are, that tropical rainfall is fueled by high values of local convective buoyancy, which depends on both the undilute buoyancy of near-surface air and on how moist is the air immediately above (*25*). These factors create a pressing need to understand how daily rainfall extremes in the Indian summer monsoon are affected by ENSO, using concomitant changes in local convective buoyancy to understand the underlying physical causes.

ENSO influences on extreme daily rainfall

Our primary metric of extreme daily rainfall is the cutoff accumulation (*26*, *27*), computed as the ratio of the variance to the mean of daily rainfall across summer days (*28*). This metric is motivated by the reliably γ-like distributions of daily rainfall (*29*): Probabilities follow a shallow power law below the cutoff before dropping off exponentially above it (Fig. 1A). Increasing the cutoff, which in the exact γ limit is identical to the scale parameter, therefore makes all extreme rain rates more likely. Our primary dataset is the India Meteorological Department (IMD) daily 0.25° by 0.25° gridded rainfall product spanning 1901 to 2020, derived from a dense, temporally varying in situ rain gauge network (*30*). We used all 122 days within each June-July-August-September (JJAS) summer season and all grid points within the "monsoonal India" domain (*31*) (Fig. 1, B and C), where the dataset is most reliable (fig. S1) (*28*). As shown below, key results are robust across alternative metrics of extremes and to another daily rainfall dataset.

Climatological summer-mean rainfall (Fig. 1B) spans from 0.4 mm to 39.4 mm day^{-1} across monsoonal India grid points and has a distinct spatial structure: high values within the southwestern coastal band, a sharp gradient moving east across the Western Ghats mountain range to much smaller values in the southeast, intermediate values in the north in most of the Central Monsoon Zone, and low values again in Rajasthan in the far northwest. This geographic pattern emerges in many properties of summer monsoon rainfall from daily to interannual timescales (*5*, *6*, *32*) (fig. S2) and includes measures of daily extremes, which additionally exhibit enhanced values in Gujarat (along the Arabian Sea coast within the Central Monsoon Zone), where most summers are quite dry but others receive synoptically driven extreme accumulations (*33*) (fig. S3).

According to lag-zero correlations of each grid point's JJAS cutoff with the standard NINO3.4 index of ENSO (*28*), extreme daily rainfall becomes more likely over much of the southwestern coastal band and central India as NINO3.4 increases but less likely over much of the southeast and far northwest (Fig. 1C). There is considerable noise, as expected for the extremes of an inherently sporadic quantity such as rainfall, but otherwise this pattern resembles the climatological summer-mean rainfall (Fig. 1B) and the established (*5*, *6*) ENSO-driven patterns in rainfall intensity.

For each of the four subregions of India outlined in Fig. 1, we computed an aggregate cutoff by combining all grid points within that region into a single daily rainfall distribution for each summer (*28*). Whereas the cutoff for Southeast India is negatively correlated with NINO3.4 and marginally statistically significant [correlation coefficient (r) = −0.16, P = 0.08], the corresponding correlations for All-Monsoonal India, the Central Monsoon Zone, and the Western Ghats regions are all significantly positive (r = +0.34, +0.34, +0.20 and P = 2×10^{-4}, 2×10^{-4}, 0.03, respectively) (*28*). Composites combining either all summers from 1901 to 2020, the 36 El Niño summers, or the 41 La Niña summers behave similarly: Except for Southeast India, the region-aggregated cutoffs are all significantly larger for the El Niño than for the La Niña composites (Fig. 2A) (*28*).

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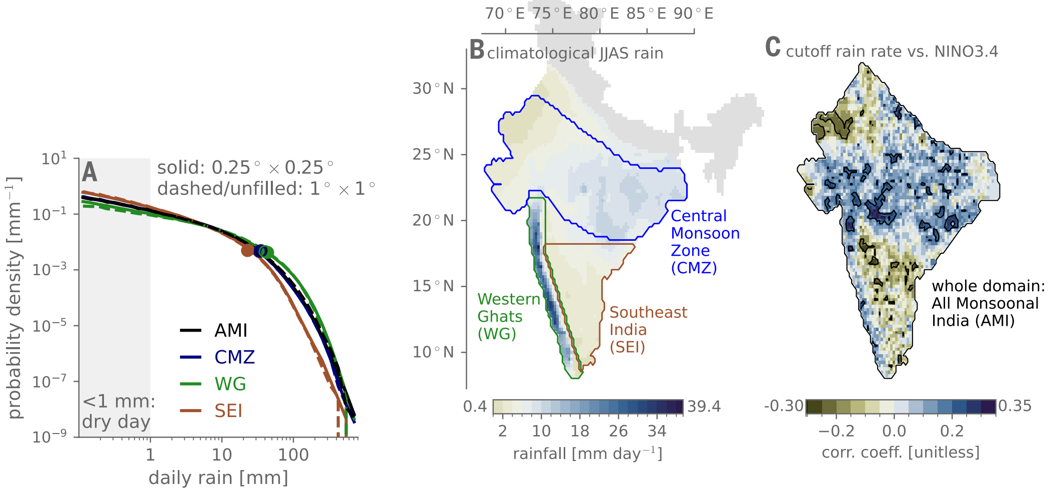

Fig. 1. Extreme daily rainfall over the Indian summer monsoon's rainiest areas becomes more likely the stronger that El Niño conditions are. (**A**) Daily rainfall empirical probability distributions, in log-log space, combining all June-July-August-September (JJAS) days from 1901 to 2020 and all grid points within all of monsoonal India (black), the Central Monsoon Zone (blue), the Western Ghats coastal band (green), or Southeast India [brown; subdomain outlines shown in (B) and (C)]. Overlaid circles indicate the climatological value of the cutoff rain rate, which is our primary metric of extreme daily rainfall. (**B**) Local climatological JJAS-mean rainfall (millimeters per day) according to the color bar. The portions of northern and northeast India shaded gray are excluded from all analyses. (**C**) Pearson correlation coefficient between detrended time series of the JJAS NINO3.4 value and each grid point's cutoff rain rate. Positive values indicate that extreme rainfall is enhanced for El Niño compared with La Niña conditions. The black contours enclose grid points with statistically significant correlations, using a false discovery rate of 0.3, which corresponds in this case to $P < 0.039$.

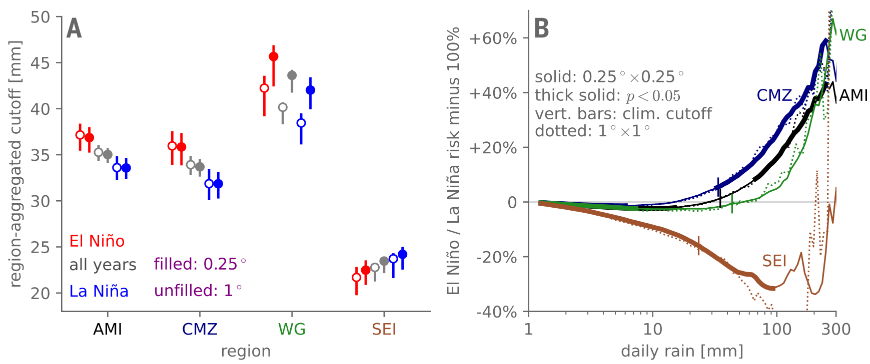

Fig. 2. Light to moderate accumulation likelihoods are suppressed everywhere, whereas heavy and extreme accumulation likelihoods are enhanced in the wet regions during El Niño summers. (**A**) Composite regional cutoffs (in millimeters) shown in filled circles. Bars indicate 95% confidence intervals based on yearly cutoff values. Open circles indicate the results computed from the IMD 1° by 1° dataset. (**B**) ENSO risk ratios: restricting to rainy days (>1 mm), the probability that rainfall exceeds a given rain rate in the El Niño composite divided by that in the La Niña composite, expressed as a percentage difference from unity. Thick lines denote where the risk ratio is significantly different from unity, according to a permutation test. Overlaid vertical lines show each region's climatological cutoff. Dotted curves are the risk ratios computed from the IMD 1° by 1° dataset.

These signals in extremes differ qualitatively from those in less-severe rain rates, which are suppressed throughout the domain by El Niño. This difference can be seen in ENSO risk ratios for each region (*26*, *27*, *29*), which are defined at each daily accumulation as the fraction of rainy days that accumulation value in El Niño summers divided by the corresponding fraction for La Niña summers (Fig. 2B) (*28*). Statistically significant reductions in the likelihoods of weak to moderate accumulations emerge in all four regions, and for Southeast India this reduction extends to the extremes as well (*28*). But for the other regions, the risk ratios minimize at moderate accumulations and then, near the climatological cutoff (overlaid vertical lines in Fig. 2B), they begin increasing quasi-exponentially with accumulation—the signature of an increased cutoff (*26*). For accumulations exceeding ~250 mm, for example, statistically significant increases in likelihoods are 43% for the whole domain and 59% for the Central Monsoon Zone; owing to greater interannual variability in this quantity for the southwestern coastal band, the 48% increase there is marginally statistically significant ($P = 0.07$). Presumably due to sampling uncertainty, behaviors for accumulations exceeding 300 mm are not statistically significant and are too erratic to be meaningfully physically interpreted. None of these key results are qualitatively sensitive to the specific methods or dataset chosen.

Correlations between NINO3.4 and pointwise exceedance counts—the number of days in a given summer exceeding a specified local-rainfall percentile or rain rate in millimeters—are predominantly negative for low through moderate values throughout the domain but become positive for sufficiently high values within the climatologically wet regions (fig. S4). Cutoffs and risk ratios estimated from γ fits to each composite distribution yield similar results (fig. S5) as do other metrics of extreme rainfall, including correlations of block maxima with

NINO3.4, pointwise Gumbel fits to the block maxima for El Niño versus La Niña composites, and quantile regressions of daily rainfall versus NINO3.4 (fig. S6) (*27*). Results are also similar in the IMD 1° by 1° gridded daily rainfall dataset (*34*), which, importantly for extremes (*35*), is generated from a fixed rather than temporally varying network of rain gauges (dashed curves and unfilled elements in Figs. 1, 2, and fig. S5) (*28*).

Lastly, whereas El Niño's propensity to generate summer-mean Indian drought weakened from the pre-satellite (1901–1978) to satellite (1979–2020) eras (*36*–*38*)—with the correlation between NINO3.4 and the all-Monsoonal India domain-mean, JJAS mean rainfall weakening by ~31%, from $P = -0.59$ to -0.41—the ENSO signals in extremes are more stationary (fig. S7) (*28*).

This finding holds in terms of the overall geographic patterns seen in pointwise cutoffs, linear regressions of the region-aggregated cutoffs on NINO3.4, and an index linearly combining the region-aggregated cutoffs of the Central Monsoon Zone, Western Ghats, and Southeast India regions (*28*). Nevertheless, within peninsular India, both the increase in extremes with NINO3.4 in the southwestern coastal band and the decrease in the southeast weaken nontrivially. As such, for the region-aggregate cutoffs, only the Central Monsoon Zone's regression against NINO3.4 is statistically significant within the satellite era. The extent to which these modest differences between epochs are physically forced versus merely sampling-driven (*37*) remains to be seen.

Mechanisms: ENSO-forced changes in convective buoyancy extremes

In general, one expects heavy rain where near-surface air is especially warm and moist and is sitting below air that is also nearly saturated. These conditions enable the near-surface air to easily begin rising, precipitating out copious amounts of water as it ascends, despite losing buoyancy to cooling through expansion and to dilution through mixing with the drier ambient air. These processes have been encapsulated into a scalar metric of convective buoyancy known as B_L (*25*, *39*), which we computed for each grid point and summer day of 1979 to 2020, using European Centre for Medium-Range Weather Forecasts Reanalysis Fifth Generation (ERA5) reanalysis data (*28*, *40*).

Daily rainfall depends strongly on local convective buoyancy, with nonlinear Spearman correlation coefficient positive at every grid point (Fig. 3A), up to +0.71, and the expected quasi-threshold relationship (*25*, *39*) in region-aggregated conditional averages (Fig. 3B) and quantiles (Fig. 3C) of rainfall as a function of local buoyancy (*28*). Rainfall is typically absent or weak until buoyancy reaches a threshold value slightly less than zero, above which rainfall increases sharply, rapidly consuming buoyancy with it (Fig. 3B). These conditional curves rise to higher rainfall values in the rainy southwestern coastal band and Central Monsoon Zone compared with those for drier southeast India, but for a given region to first approximation they are insensitive to the state of ENSO (Fig. 3C).

This consistency of the rainfall-buoyancy relationships across ENSO states, combined with the responses of the convective buoyancy distributions themselves to ENSO, help explain the responses of rainfall to ENSO. At the summer-mean timescale, pointwise correlations of convective buoyancy against NINO3.4 are negative at every grid point (fig. S8). But for extremes, such as 99th-percentile rainy-day values, they are significantly positive over much of central-eastern India but significantly negative over much of the southeast and northwest (Fig. 4A). These signals are shared by the individual buoyancy components (Fig. 4, B and C): Both nearly saturated and very gravitationally unstable days are predominantly enhanced over eastern-central India versus suppressed over the northwest and southeast.

What, then, accounts for these convective buoyancy signals? Preliminary results point toward synoptic low-pressure systems (*41*), which generate heavy rainfall within their cores (*42*) and track predominantly over Central India, although occasionally traveling farther

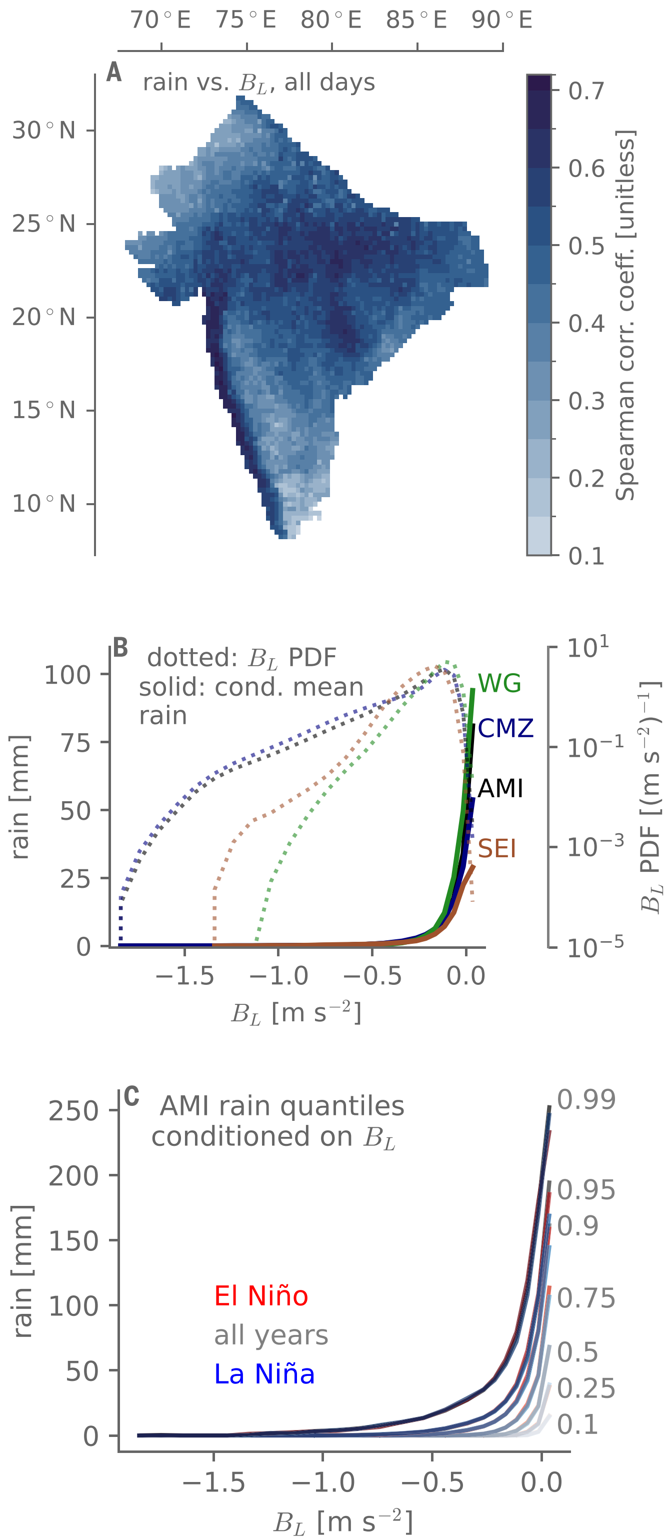

Fig. 3. Local convective buoyancy strongly controls daily rainfall. (**A**) Spearman correlation coefficient across all JJAS days from 1979 to 2020 at each grid point between daily rainfall and a metric of convective buoyancy (B_L) computed from ERA5 reanalysis data. (**B**) In dotted curves, probability density functions of B_L for each region, according to the right axis. In solid curves, daily rainfall is conditionally averaged on the buoyancy metric, according to the right axis. (**C**) For the entire monsoonal India domain, composite quantiles of rainfall are conditioned on B_L as labeled. For each quantile, red curves for El Niño are almost entirely obscured by others because of weak ENSO dependence in the relationships.

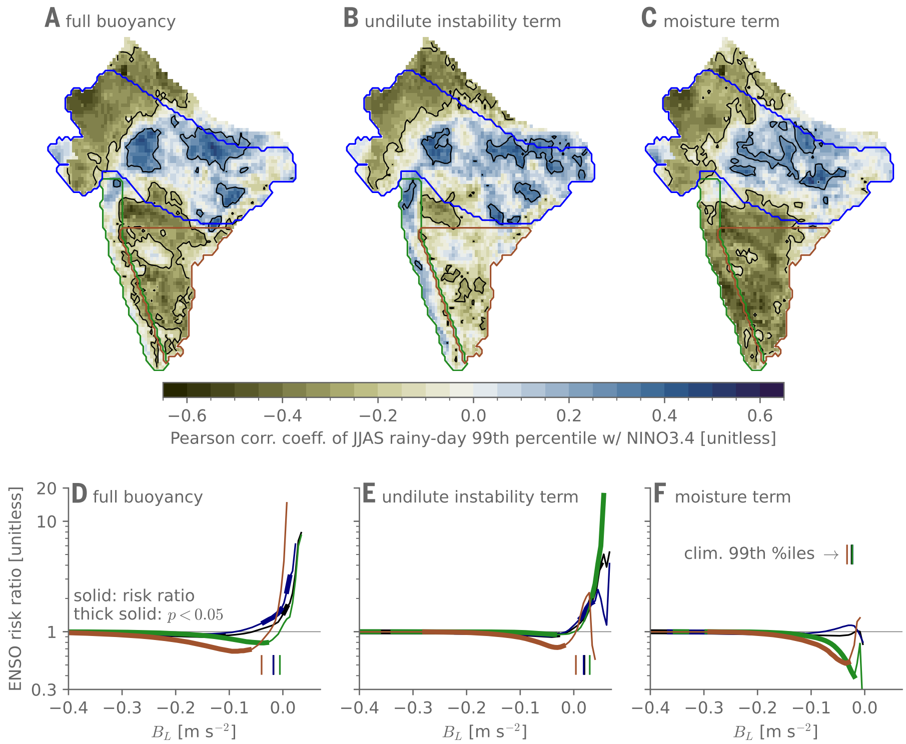

Fig. 4. Right-tail daily convective buoyancy occurrence under ENSO largely tracks with the extreme rainfall. (**A** to **C**) Pearson correlation coefficient of NINO3.4 versus 99th-percentile daily values of (A) the full buoyancy metric, (B) just the undilute instability term, and (C) just the moisture term, for each JJAS. The black contours enclose areas deemed statistically significant on the basis of a false discovery rate of 0.3. (**D** to **F**) Regional ENSO risk ratios of rainy-day buoyancy, for (D) the full metric, (E) just the undilute instability term, or (F) just the moisture term. Thick lines denote where the risk ratio is significantly different from unity, on the basis of a permutation test. Overlaid vertical lines denote the climatological 99th-percentile values for each region of the given buoyancy component.

south or extending to Rajasthan in the northwest (*33*). A prior study indicates that the cores of monsoon low-pressure systems are very humid and that during El Niño summers, the rainfall within those cores is enhanced (*43*). One possibility is that those low-pressure system tracks over and ending in central India are favored by El Niño at the expense of those over peninsular India and any long tracks that might reach northwestern India. This would promote very humid, buoyant days—and with them extreme rainfall likelihoods—in central-eastern India and suppress them in the southeast and northwest.

For the southwestern coastal band—which is less directly influenced by these synoptic systems—whereas nearly saturated days above the boundary layer are strongly suppressed in El Niño summers, highly gravitationally unstable days are strongly enhanced. This is particularly evident in risk ratios of the convective buoyancy terms on rainy days (Fig. 4, D to F) (*28*): Gravitationally unstable days are as much as 17 times more likely in El Niño than La Niña summers. Moreover, according to differences in the ENSO composites of rainfall conditioned on the buoyancy terms (fig. S9), tropospheric dryness inhibits heavy rainfall somewhat less during El Niño summers. We therefore speculate that the anomalous time-mean free-tropospheric subsidence and dryness under El Niño (fig. S10) suppress moderate rain events, enabling higher undilute instability values to occasionally build up, fueling extreme convection once a passing wave or advective event remoistens the lower free troposphere just enough. This mechanism would be broadly consistent with observational and modeling evidence for less frequent but more intense rain events in drier ambient conditions (*44*), such as in the pre-monsoon season (*45*)

Outlook: Seasonal forecasts, climate change, and other extensions

Despite ENSO's known "spring predictability barrier" (*46*), existing seasonal forecasts of ENSO during boreal summer are skillful (*47*); for example, the multimodel-mean forecasted JJAS value of NINO3.4 from the North American Multi-Model Ensemble (NMME) (*48*) seasonal forecast integrations initialized annually on 1 May are correlated with the actual value at $r = 0.86$ over 1981–2020 (*28*). We have recomputed the correlations between the JJAS cutoff rain rate and NINO3.4 using this forecasted value, and the overall signals remain (fig. S11). This raises the prospect of skillful ENSO-based seasonal forecasts of extreme rainfall probabilities within monsoonal India. However, the statistical significance weakens nontrivially, and risk ratios that use NMME 1-May forecasted JJAS NINO3.4 show considerably weaker separation between El Niño and La Niña summers. Providing societally actionable information may therefore require even more-skillful forecasts of the state of ENSO.

The observed long-term trend toward more extreme daily rainfall events in central India into the early 2000s has been argued to stem from mean moistening of the troposphere due to global warming (*17*, *18*) as have future projections of increased extreme rainfall events globally (*27*). Notwithstanding that more recent trends in the Indian summer monsoon have been flat or negative (*20*), underlying this argument are the assumptions that right-tail daily moisture events (i) track with the season-mean value and (ii) predominate over changes in undilute instability. On the basis of our results, however, for ENSO, (i) is violated within central India, and (ii) is violated within the southwestern coastal band—precisely the regions where daily extreme rainfall is projected to increase most in model simulations under future global warming (*49*). This finding argues for revisiting both observed trends and model projections of extreme rainfall, incorporating the full daily distributions of rainfall and of the convective buoyancy terms. This evaluation must be done in tandem with careful analysis of secular changes in monsoon low-pressure systems—both weaker lows and stronger depressions—because projected increases in rainfall intensity within these systems likely play a major factor in the overall expected increase in Indian summer extreme rainfall (*50*). Only then can we meaningfully evaluate whether these observed, interannual, internally generated ENSO signals constitute a meaningful emergent constraint on model-projected, secular, externally forced trends (*51*).

Our approach could also be applied to climate variability modes in the Indian (*52*, *53*) and Atlantic Oceans (*54*) thought to influence the summer monsoon, as well as to hourly rainfall. Many Indian summer monsoon rain events are considerably shorter than 24 hours in duration (*55*), and a study (*15*) using hourly rainfall suggests that the average rain rate during rainy hours increases with El Niño over much of the Tropics. Perhaps, as for daily rain in the Indian summer monsoon, these signals reflect buoyancy-driven increased extremes.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S11; References (*56*–*66*)

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